

Methods Appendix:

Estimating Local LGBTQ Populations

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Overview

I estimate the expected LGBTQ population of every U.S. Census tract in five conceptual stages: (1) measure LGBTQ identification and gender-identity composition in Household Pulse Survey (HPS) microdata, softly calibrated to external Gallup benchmarks; (2) stabilize these rates against sampling noise using two complementary smoothing techniques — one across nearby ages, one across states; (3) project the resulting rates onto Census tracts using ACS 5-year age–sex population counts; (4) reweight the within-state spatial distribution using the observed geographic concentration of same-sex couples as a proxy for the broader LGBTQ population, smoothed across space so the result doesn’t show artificial jumps at PUMA boundaries; and (5) enforce a capacity constraint so no tract’s estimate can exceed its underlying population. Each stage is detailed in its own section below, with the precise notation defined immediately below.

Notation

Throughout, i indexes survey respondents and w_i denotes HPS survey weights. I fix notation of gender and sex up front:

- **Sex assigned at birth**, $s \in \{M, F\}$. Since the ACS does not ask about gender identity, I begin with sex assigned at birth. Table B01001 provides tract-level counts of age-sex cells.
- **Gender identity**, $g \in \{\text{cis}, \text{nb}, \text{trans}\}$, estimated from HPS in Section 4. This is *not* the same as the final output gender bucket used for population accounting on the map. Instead, the gender identity variable simply flags an individual’s relationship to their sex assigned at birth. For example, transgender respondent’s gender identity is not their sex assigned at birth, so “trans” respondents with $s = M$ are trans women and those with $s = F$ are trans men.

I therefore define a third symbol, the **gender bucket** $k \in \{m, w, \text{nb}\}$ (man, woman, non-

binary), combining the sex assigned at birth and gender identity:

$$k(s, g) = \begin{cases} m & (s = M, g = \text{cis}) \text{ or } (s = F, g = \text{trans}) \\ w & (s = F, g = \text{cis}) \text{ or } (s = M, g = \text{trans}) \\ \text{nb} & g = \text{non-binary} \end{cases}$$

In words: a cisgender respondent’s bucket matches their sex assigned at birth; a transgender respondent’s bucket does not match their sex assigned at birth. Non-binary respondents form their own bucket regardless of s . Sections 6–8 report quantities indexed by k ; Section 4 reports the intermediate quantities indexed by g that k is built from.

The remaining recurring symbols are: a for exact age or an ACS age bin; $\ell \in \{\text{LG}, \text{Bisexual}, \text{Queer}\}$ for LGBT subcategory (excluding Trans, which is cross-cut with k rather than treated as a fourth member of ℓ); p for a PUMA (Public Use Microdata Area); and b for a Census tract.

1 Measurement in HPS Microdata

Let $Y_i \in \{0, 1\}$ indicate whether individual i identifies as LGBTQ. In the HPS, I mark respondents as LGBTQ if they report a sexual orientation besides straight, a trans gender identity, or both. Note that non-binary people are not automatically LGBTQ, though an overwhelming majority report a non-straight sexuality too. Sex assigned at birth s_i and age a_i are taken directly from HPS (age top-coded at 62 to limit disclosure risk in the sparse cells of older cohorts). At this raw stage, everything is purely survey-based and reflects HPS’s own sampling variability – the calibration and smoothing steps that follow exist precisely to tame that variability without discarding real geographic and demographic heterogeneity.

Survey weights w_i are retained throughout all of the estimation steps below.

2 Soft Calibration to Gallup Marginals

HPS and Gallup differ in survey mode, sampling variation, and field timing, so I don’t expect them to agree exactly and I don’t force them to. Instead, HPS-based probabilities are *softly* calibrated toward Gallup national benchmarks, nudged toward external validity.

Each individual begins with a baseline probability $p_{0i} = \text{clipped}(Y_i)$ (clipped away from exactly 0 or 1 so the subsequent logit transform is well-defined). A logit adjustment model is estimated:

$$\text{logit}(p_i) = \text{logit}(p_{0i}) + X_i\theta,$$

where X_i includes generation effects and generation-by-sex interactions. The parameter vector θ minimizes a weighted objective combining (i) a deviation penalty that keeps estimates close to HPS baselines, and (ii) a margin penalty that discourages deviation from Gallup targets, optimized via BFGS. In practice, this keeps the vast majority of state-level estimates within one to two percentage points of Gallup’s national top-line estimate of roughly 9 percent, while still letting genuine state-level heterogeneity show through.

3 State–Sex–Age LGBT Rate Estimation

Using calibrated probabilities, sex–age–state LGBT identification rates are computed as a straightforward weighted mean:

$$\hat{P}(Y = 1 \mid s, a, \text{state}) = \frac{\sum_i w_i p_i^{\text{adj}}}{\sum_i w_i},$$

with effective sample size $n_{\text{eff}} = (\sum_i w_i)^2 / \sum_i w_i^2$ tracking how much genuine information backs each cell. Weekly HPS survey waves are pooled and averaged to reduce short-run noise before this step.

4 Age-Local Dirichlet Smoothing of Gender Composition

This step estimates $P(g \mid \ell, s, a)$ – the distribution of gender-identity type $g \in \{\text{cis}, \text{nb}, \text{trans}\}$ within each LGBT subcategory, sex assigned at birth, and exact age (see *Notation* above for why g is deliberately kept distinct from the final output bucket k).

The raw cell rate is

$$\hat{p}_{g,\ell,s,a} = \frac{n_{g,\ell,s,a}}{\sum_g n_{g,\ell,s,a}}, \quad n_{\text{eff},\ell,s,a} = \frac{(\sum w_i)^2}{\sum w_i^2},$$

where $n_{g,\ell,s,a}$ denotes survey-weighted counts. Some cells are thin; in particular, non-binary and transgender respondents at a single exact age within a single sexuality category might often yield sparse cells. Thin cells lead add to the raw shares. I address this with two smoothing passes: first borrowing strength from *nearby ages* within a state (this section), then, in Section 5, borrowing strength across *states* for the coarser LGBT subcategory shares.

4.1 Age-Local Prior Construction

I motivate the smoothing with a concrete pattern in the data: younger AFAB (assigned female at birth) lesbians are roughly 15 percentage points more likely to identify as non-binary than AMAB gay men in the same age cohort. I also don’t want a single noisy exact-age cell to dominate the estimate. Age-local smoothing does exactly this: it borrows information from *nearby* ages, weighted by how much data backs each neighbor and how close in age they are.

For a focal age a , neighboring ages $a' \neq a$ receive weights

$$w_{a'}^{(a)} = n_{\text{eff},\ell,s,a'} \exp\left(-\frac{|a - a'|}{\tau}\right),$$

where $\tau > 0$ controls how quickly influence decays with age distance. The age-local prior mean is then the weighted average of neighboring rates:

$$m_g^{(a)} = \frac{\sum_{a' \neq a} w_{a'}^{(a)} \hat{p}_{g,\ell,s,a'}}{\sum_{a' \neq a} w_{a'}^{(a)}}.$$

4.2 Dirichlet Moment Matching

A Dirichlet prior is centered at $\mathbf{m}^{(a)}$, with concentration parameter estimated via method of moments:

$$\alpha_0 = \text{mean}_g \left(\frac{m_g(1 - m_g)}{v_g} - 1 \right),$$

where v_g is the weighted variance of $\hat{p}_{g,\ell,s,a'}$ across neighboring ages a' – intuitively, how much the neighbors agree with each other. The prior vector is $\alpha_g = \alpha_0 m_g$.

4.3 Posterior Shrinkage

The smoothed estimate for focal age a blends the raw cell rate with the age-local prior, weighted by how much real data each side has:

$$\tilde{p}_{g,\ell,s,a} = \frac{n_{\text{eff},\ell,s,a} \hat{p}_{g,\ell,s,a} + \alpha_g}{n_{\text{eff},\ell,s,a} + \alpha_0}.$$

A well-sampled cell (n_{eff} large relative to α_0) stays close to its raw rate; a thin cell shrinks toward the age-local prior.

5 Cross-State Shrinkage of LGBT Subcategory Shares

The same thin-cell problem recurs one level up: some LGBT subcategory shares are sparsely measured within individual states (transgender respondents in any single state, or bisexual respondents in a small state like Wyoming). For each sex s , age bin a , and LGBT category ℓ , I define the state-level share

$$\hat{\pi}_{\ell,s,a,j} = P(\ell \mid Y = 1, s, a, \text{state } j),$$

with effective sample size $n_{\text{eff},j}$. Rather than age, the natural borrowing dimension here is *across states*: a beta prior is fit by method of moments across the 50-state (plus DC) distribution of $\hat{\pi}_{\ell,s,a,j}$, $\pi_{\ell,s,a,j} \sim \text{Beta}(\alpha, \beta)$, and posterior shrinkage gives

$$\tilde{\pi}_{\ell,s,a,j} = \frac{n_{\text{eff},j} \hat{\pi}_{\ell,s,a,j} + \alpha}{n_{\text{eff},j} + \alpha + \beta}.$$

Final shares are renormalized within each state–sex–age bin so they sum to one. This preserves genuine state-to-state heterogeneity – some states really do have different LGBT subcategory compositions – while damping the noise that would otherwise dominate small states or thin cells.

6 ACS Projection to Census Tracts

Let $N_b^{(s,a)}$ denote the ACS 5-year population count for sex assigned at birth $s \in \{M, F\}$ and age bin a in Census tract b . I now combine every rate estimated above – identification, subcategory share, and gender-identity composition – with local ACS demographics to get an expected count

for each tract, LGBT subcategory, and output gender bucket $k \in \{m, w, nb\}$:

$$\hat{E}[\text{count}_{k,\ell,b}] = \sum_{s,a} N_b^{(s,a)} \cdot \hat{P}(Y = 1 \mid s, a, \text{state}) \cdot \tilde{\pi}_{\ell,s,a} \cdot \sum_{g: k(s,g)=k} \tilde{p}_{g,\ell,s,a}.$$

The innermost sum collapses the gender-identity-type rate $\tilde{p}_{g,\ell,s,a}$ from Section 4 into the output bucket k using the $(s, g) \mapsto k$ mapping defined in *Notation*. Because identification, subcategory, and gender-identity rates all vary with age and sex, and both vary locally, expected LGBTQ counts vary across tracts even within a single state – this section captures only the *demographic composition* channel of that variation; the next section adds a *geographic clustering* channel on top of it.

Tracts that span PUMA boundaries receive population-weighted contributions from each overlapping PUMA, proportional to the area of intersection.

7 PUMA Reweighting

Spatial sorting shapes the spatial distribution of LGBTQ people across United States. LGBTQ cluster for many reasons, including proximity to community infrastructure, local social tolerance, idiosyncratic amenity preferences, and more. Yet the projection in Section 6 allocates LGBTQ population using only local age and sex composition – it offers no mechanism for the spatial sort of LGBTQ people beyond age and sex. I use the observed concentration of same-sex couples as a proxy for the broader LGBTQ population’s spatial distribution.

Formally, within a state I assume same-sex couples and all LGBTQ people have the same probability of selecting a given PUMA conditional on living in that state. This is an approximation, not an identity: same-sex couples tend to be older, wealthier, better-educated, and more likely to identify as gay or lesbian than the broader LGBTQ population, so their spatial concentration could systematically differ from, say, young queer individuals’. I accept this approximation because it still captures real geographic signal the age/sex projection alone misses entirely, and because the calibration strength is a single tunable parameter (γ below) rather than a hard constraint.

7.1 PUMA Calibration Factors

For each PUMA p , define the uncalibrated LGBTQ share of the state total projected in Section 6:

$$S_p = \frac{\sum_{b \in p} \hat{E}[\text{LGBTQ}_b]}{\sum_{p'} \sum_{b \in p'} \hat{E}[\text{LGBTQ}_b]}.$$

I construct a same-sex-couple signal from ACS Table B09019 counts of same-sex (SS) and opposite-sex (OS) couples by PUMA. PUMAs with fewer than 100 total couples are dropped as too noisy to trust, and the raw same-sex share is winsorized to the 1st–99th percentile across PUMAs before transforming, to keep a handful of tiny, extreme PUMAs from dominating the calibration:

$$\hat{q}_p = \text{winsorize}_{[0.01, 0.99]} \left(\frac{\text{SS couples}_p}{\text{SS couples}_p + \text{OS couples}_p} \right).$$

The signal is then logit-transformed and mean-centered relative to the *uncalibrated* LGBTQ distribution, so it measures relative rather than absolute concentration:

$$\tilde{\sigma}_p = \text{logit}(\hat{q}_p) - \sum_{p'} S_{p'} \text{logit}(\hat{q}_{p'}).$$

For PUMAs with fewer than 100 couples (dropped above, hence no $\hat{q}_p$), I set $\tilde{\sigma}_p = 0$: no tilt in either direction, equivalent to leaving the raw uncalibrated share S_p untouched.

The tilt factor for PUMA p is $T_p = \exp(\gamma \tilde{\sigma}_p)$, where $\gamma > 0$ governs how strongly the reweighting responds to the same-sex-couple signal (site defaults to $\gamma = 0.5$, a deliberately conservative choice given the approximation discussed above). The calibrated target share and resulting calibration factor are

$$S_p^* = \frac{S_p T_p}{\sum_{p'} S_{p'} T_{p'}}, \quad \hat{c}_p = \frac{S_p^*}{S_p}.$$

7.2 Inverse-Distance-Weighted Smoothing to Tract Level

Applying $\hat{c}_p$ uniformly to every tract within PUMA p produces a step-function surface, with hard discontinuities at PUMA boundaries. Yet those jumps stem from where the Census Bureau happened to draw PUMA lines, not real demographic transitions. Hence, I smooth the calibration factors to the tract level using inverse-distance weighting (IDW) instead of applying them as a step function.

Let $\mathbf{x}_b$ and $\mathbf{x}_p$ denote the projected-coordinate (EPSG:3857) centroids of tract b and PUMA p . The IDW-smoothed calibration factor for tract b is

$$\tilde{c}_b = \frac{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2} \hat{c}_p}{\sum_p \|\mathbf{x}_b - \mathbf{x}_p\|^{-2}}.$$

A tract deep inside a single PUMA receives a factor close to that PUMA's $\hat{c}_p$; a tract near a PUMA boundary receives a blend of the neighboring PUMAs' factors, weighted by inverse squared distance. This preserves the geographic heterogeneity the calibration is meant to capture while producing a spatially continuous surface with no visible seams. Checked against Michigan's 4,148 tracts and 68 PUMAs, IDW weights sum to 1.0 in every tract by construction, and resulting calibration factors range from 0.81 to 1.19 – a meaningfully wide spread of local tilt, but nowhere near a discontinuous jump.

All subcategory estimates in tract b are multiplied by $\tilde{c}_b$, and the whole state is rescaled so totals match the uncalibrated projection from Section 6:

$$\text{scale} = \frac{\sum_b \hat{E}[\text{LGBTQ}_b]}{\sum_b \tilde{c}_b \hat{E}[\text{LGBTQ}_b]}.$$

8 Capacity Constraint

The PUMA reweighting above is, by design, a multiplicative tilt – and multiplicative tilts have no natural ceiling. In high-concentration areas, reweighting and state-level rescaling can occasionally push a tract’s estimated count for an output gender bucket k above that bucket’s actual population, which is nonsensical: no tract can contain more LGBTQ men than it contains men.

For output gender bucket $k \in \{m, w, nb\}$ and tract b , let $\hat{N}_{k,b}^{\text{LGBTQ}}$ denote the post-calibration LGBTQ count and $N_{k,b}$ the bucket’s total population. I cap each bucket at its population ceiling and scale every LGBTQ subcategory within that bucket down proportionally, preserving relative composition:

$$\kappa_{k,b} = \min\left(1, \frac{N_{k,b}}{\hat{N}_{k,b}^{\text{LGBTQ}}}\right),$$

with every subcategory count for bucket k in tract b multiplied by $\kappa_{k,b}$. Imposition of the capacity constraint is rare; in Michigan, the constraint bound only 64 tracts. Even when the constraint affects a tract’s population, the aggregate consequences of the constraint are minute.